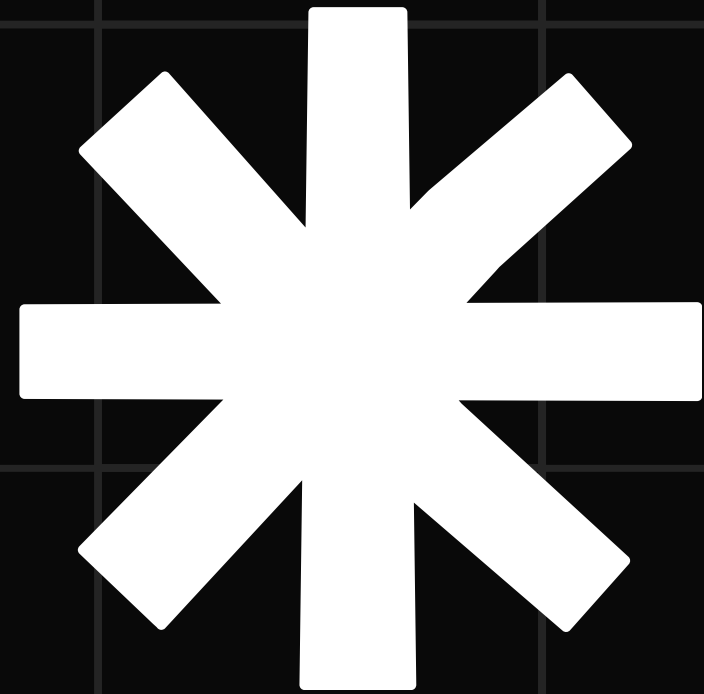


50.021



Real vs AI Generated Images

Yong Jun Han (1009942)

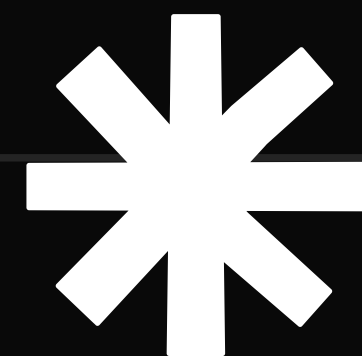


Contents

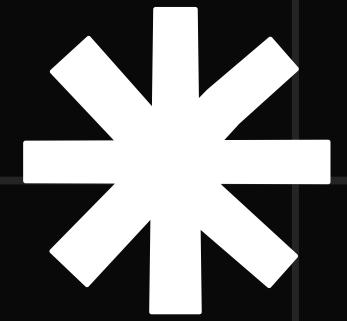
- Why the Need to Differentiate?
- Dataset to be Used
- Existing Works
- Preliminary Ideation

AI image generators are easily accessible to the public today. From Midjourney to Deepfakes,

These images can be used to spread a false narrative, or to swindle.



Case Study



The “Brad Pitt” Incident

A French woman was scammed by a fraudster using AI generated images.

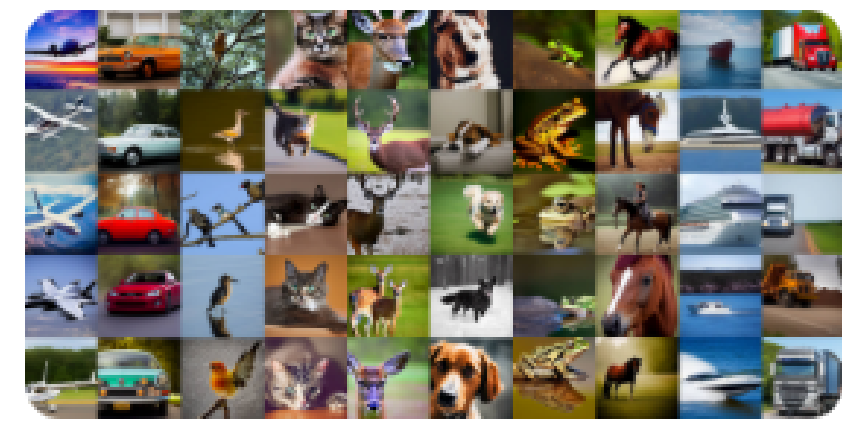
The woman was contacted by someone claiming to be Brad Pitt’s mother, and later by someone claiming to be the actor.

“Brad Pitt” then sent her photos of him in a hospital bed, successfully duping her.

Aftermath: £700,000 and a broken marriage

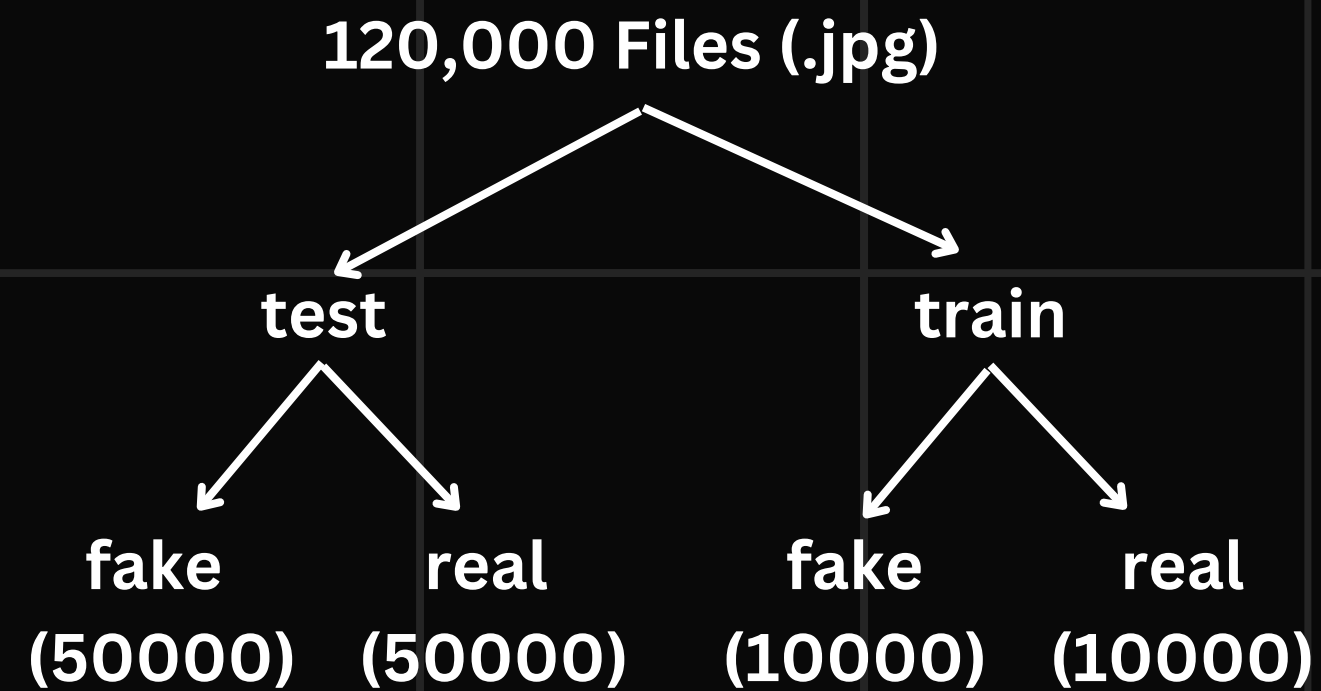


CIFAKE: Real and AI-Generated Synthetic Images

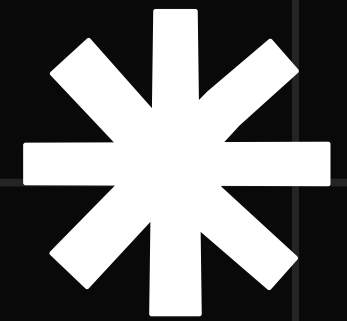


Dataset

CIFAKE: Real and AI-Generated Synthetic Images (Kaggle)



Existing Works

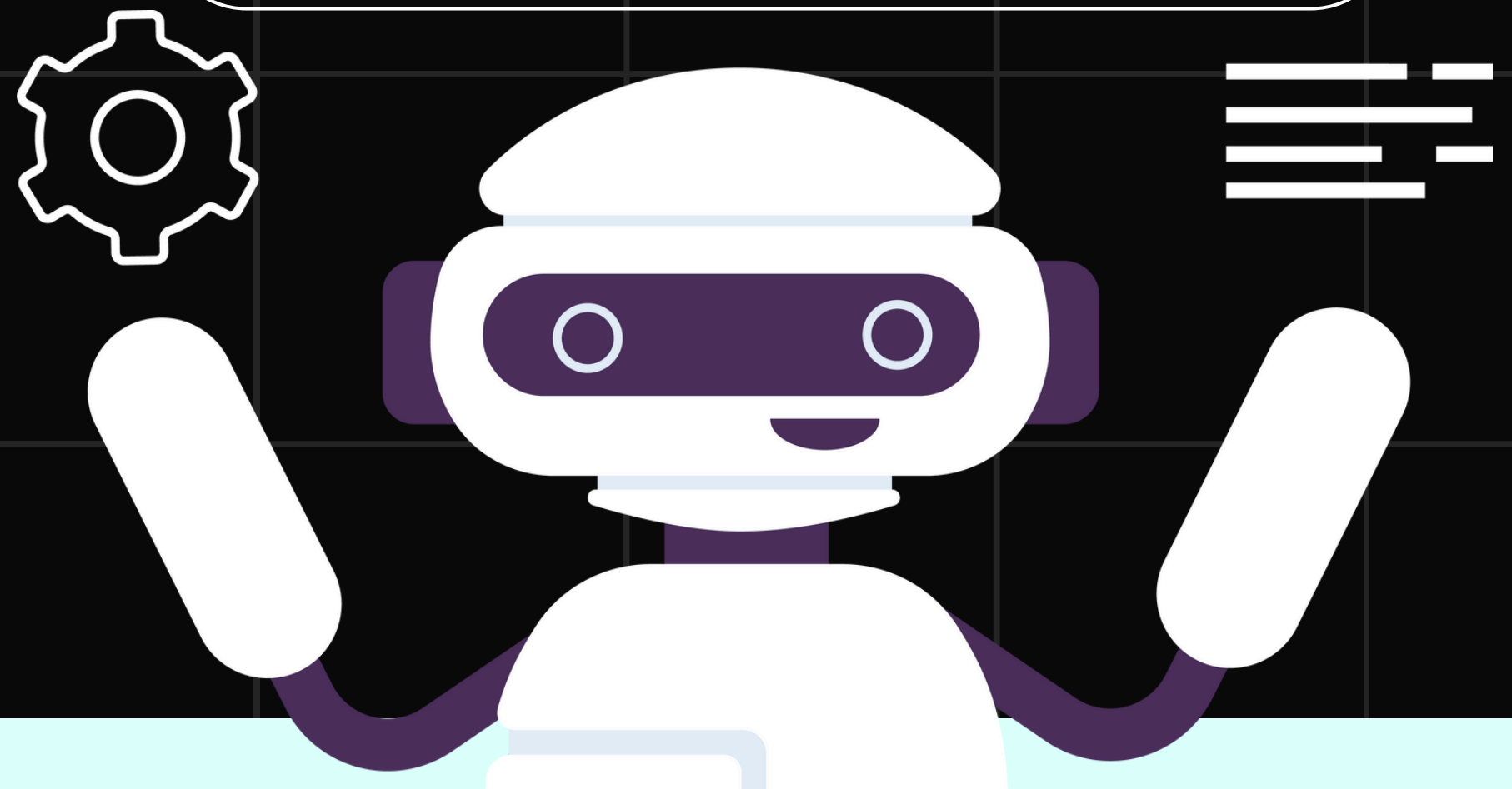
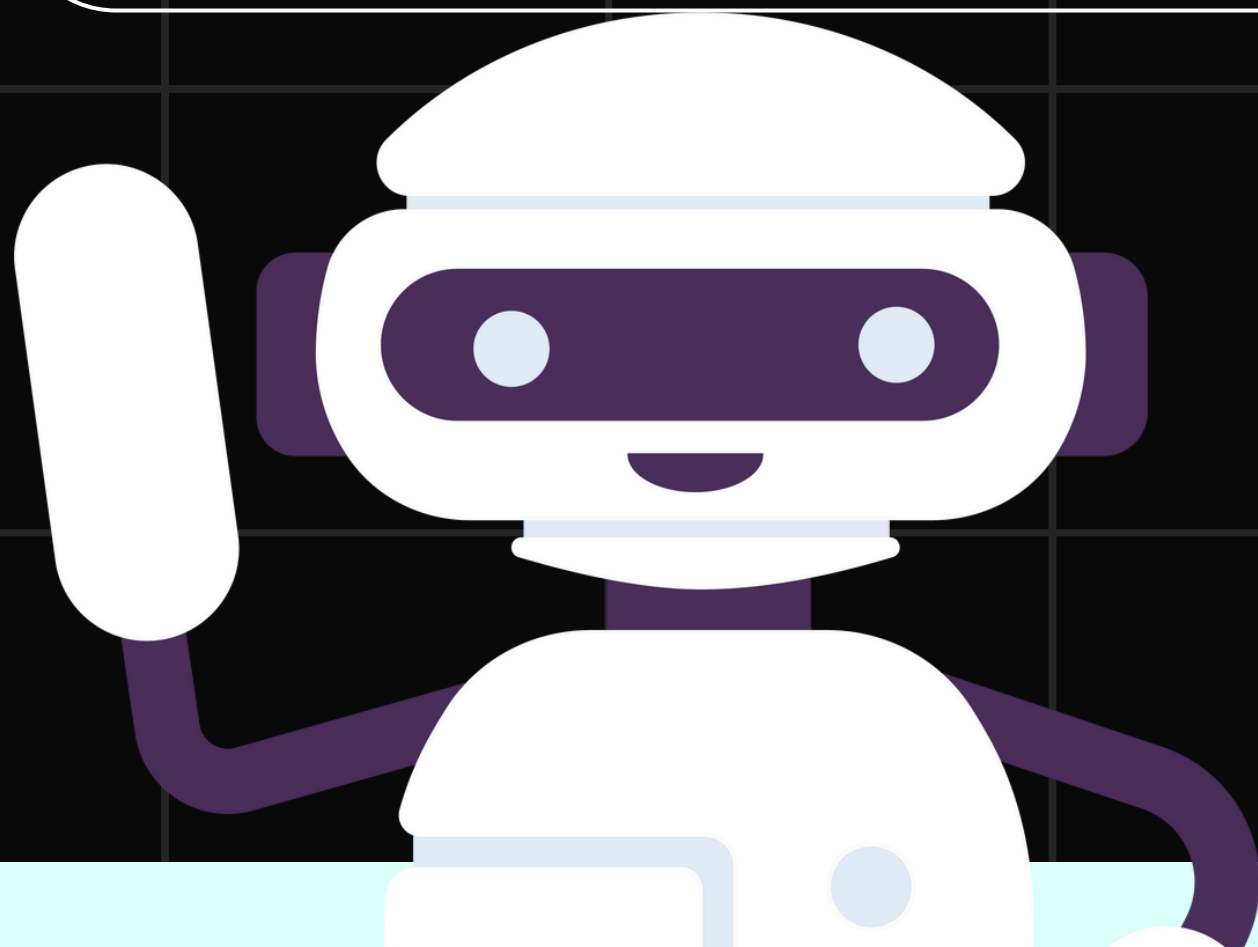


Convolved Neural Networks (CNN)

- Supervised Learning.
- Deep NN utilising convolutional layers.

Generative Adversarial Networks (GAN)

- Unsupervised learning.
- Discriminative network evaluates candidates generated by the generative network.



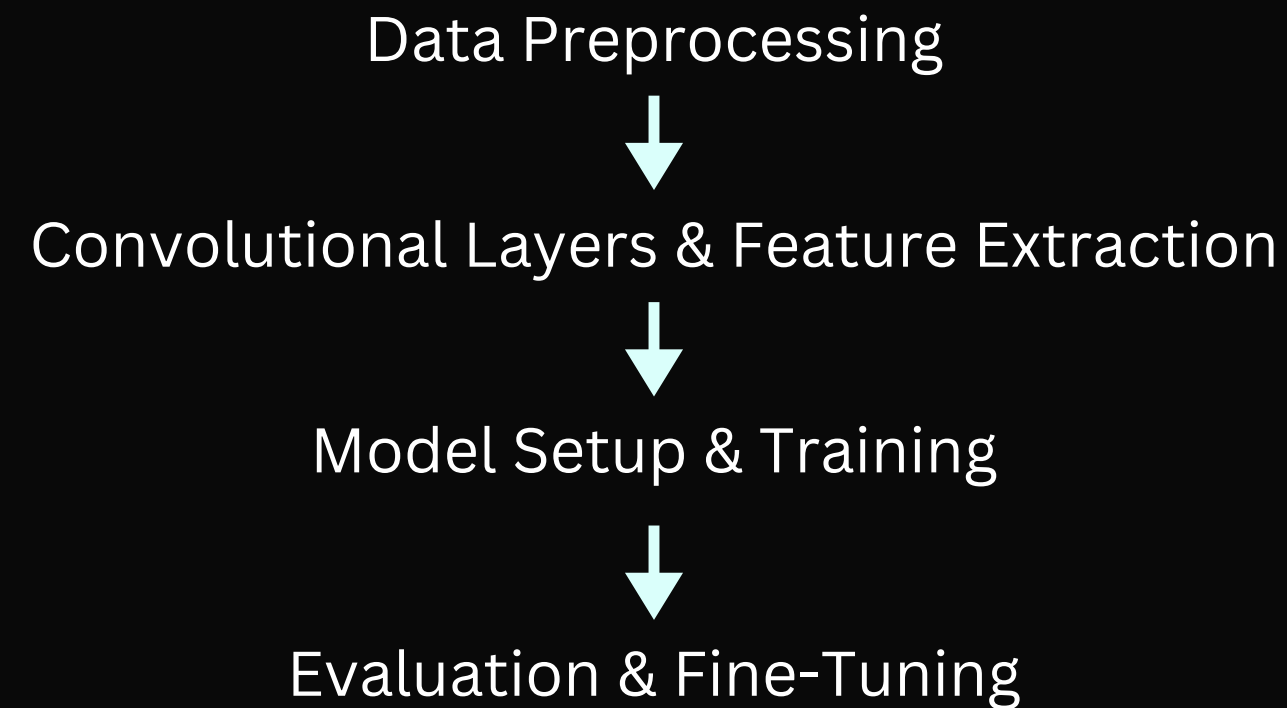
Preliminary Ideation

Preprocessing

- The train dataset can be preprocessed to increase model's accuracy.
- Data augmentation can increase variability by rotating, flipping or adding noise to the images.
- Images may not be in a standard format, so resizing is important.

Preliminary Ideation

Model to be Used: CNN



Thank you!

